CPEN 416

Lecture 17: manipulating density matrices; quantum channels

Thursday 12 November 2026

Announcements

- Remember to fill out weekly project surveys
- Tutorial Monday: **change of plans**, practice for observables / expectation values; next tutorial assignment week after
- Quiz 8 Tuesday

Last time

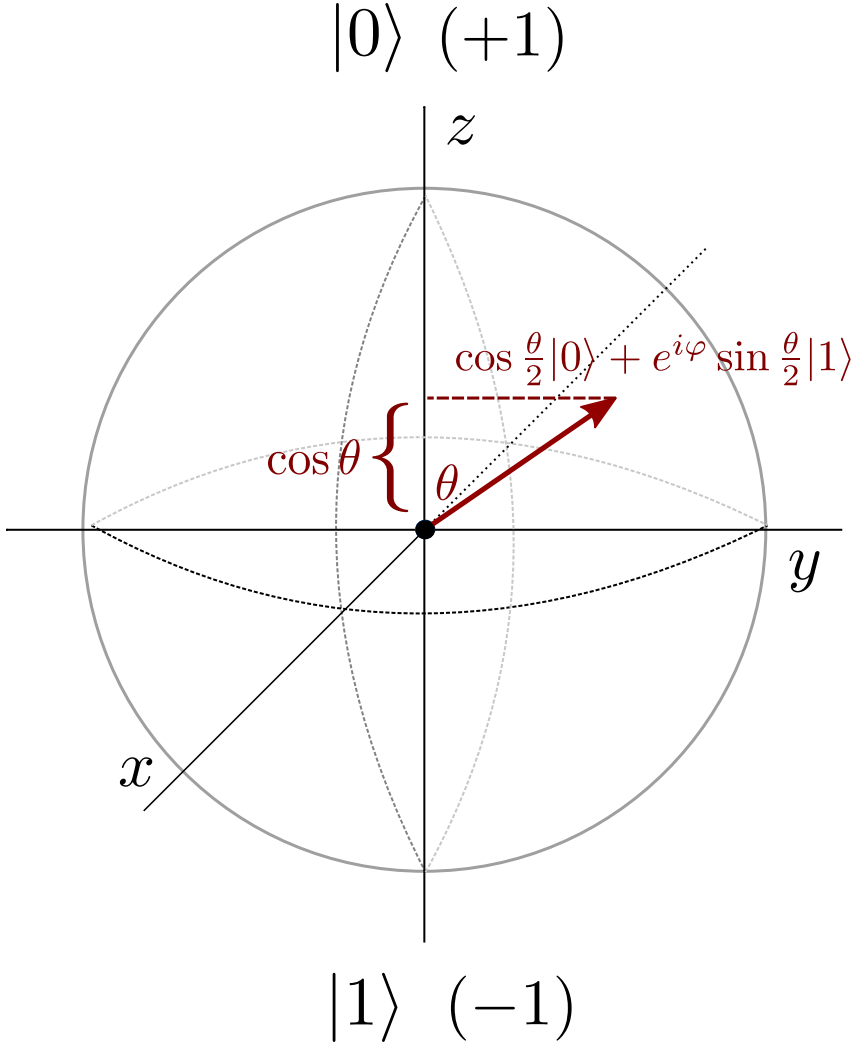
We introduced observables

We measure an observable by associating its eigenvectors with a basis and assigning real values (its eigenvalues) as outcomes.

The *expectation value* of an observable is

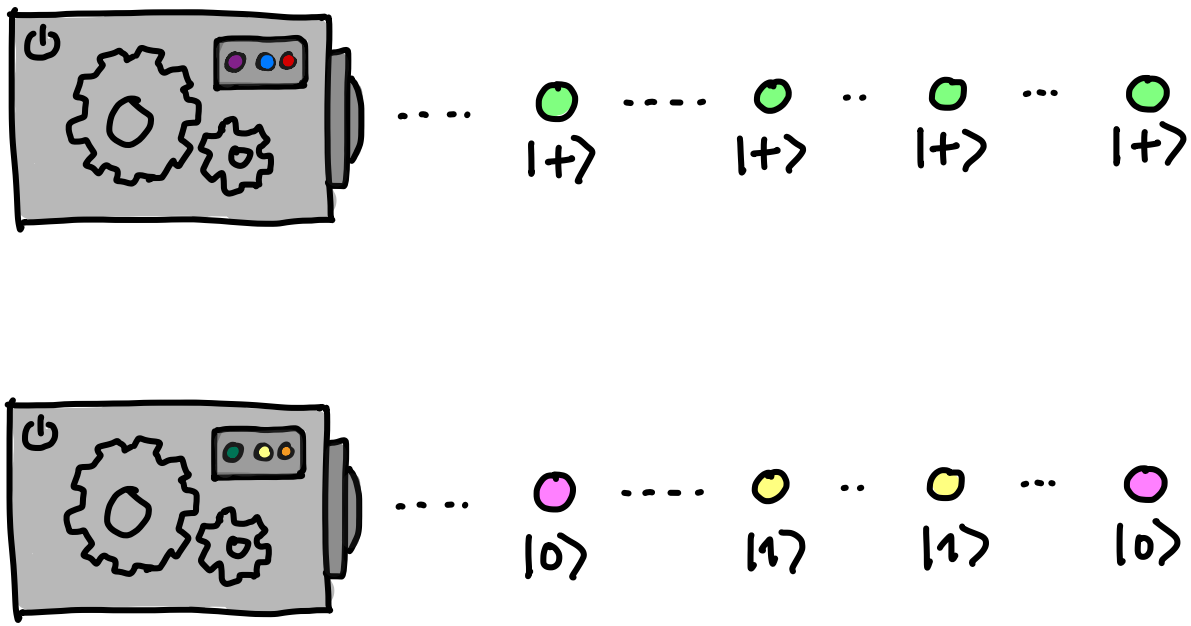
Last time

We associated expectation values of Paulis to the Bloch vector.



Last time

We introduced *mixed states*.



Density matrices

Mixed states are represented by density matrices,

Important properties of density matrices:

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Learning outcomes

- Apply unitary operations to mixed states
- Perform measurements on mixed states
- Define and apply quantum channels to quantum states

Unitary operations and mixed states

Review exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

Unitary operations and mixed states

Exercise: if H to our mixed state ($|+\rangle$ w/prob. $1/3$, $|0\rangle$ w/prob. $2/3$), what is the new state?

Unitary operations and mixed states

For pure state $|\psi\rangle$ and operation U,

As mixed states,

More generally,

Mixed states and measurements

Recall that for a pure state $|\psi\rangle$, the probability of measuring and observing it in state $|\varphi\rangle$ is

We can rewrite this...

$\Pi_\varphi = |\varphi\rangle\langle\varphi|$ is the density matrix of $|\varphi\rangle$, which is a *projector*.

Mixed states and measurements

For mixed states, measurement follows the **Born rule**:

where $\{\Pi_i\}$ is a **positive operator-valued measure (POVM)**.

The POVM elements satisfy

Exercise: show this is equivalent to a projective measurement for pure states.

Mixed states and measurements

Example: $\{|+\rangle\langle+|, |-\rangle\langle-|\}$.

Given a state ρ,

Exercise: Show that $\{|+\rangle\langle+|, |-\rangle\langle-|\}$ form a legit POVM.

Mixed states and measurements

Exercise: Suppose we prepare our system in $|+\rangle$ with probability $1/3$ and $|0\rangle$ with probability $2/3$. What is the probability of obtaining the POVM outcome $\Pi_+ = |+\rangle\langle+|$?

Mixed states and the Bloch sphere

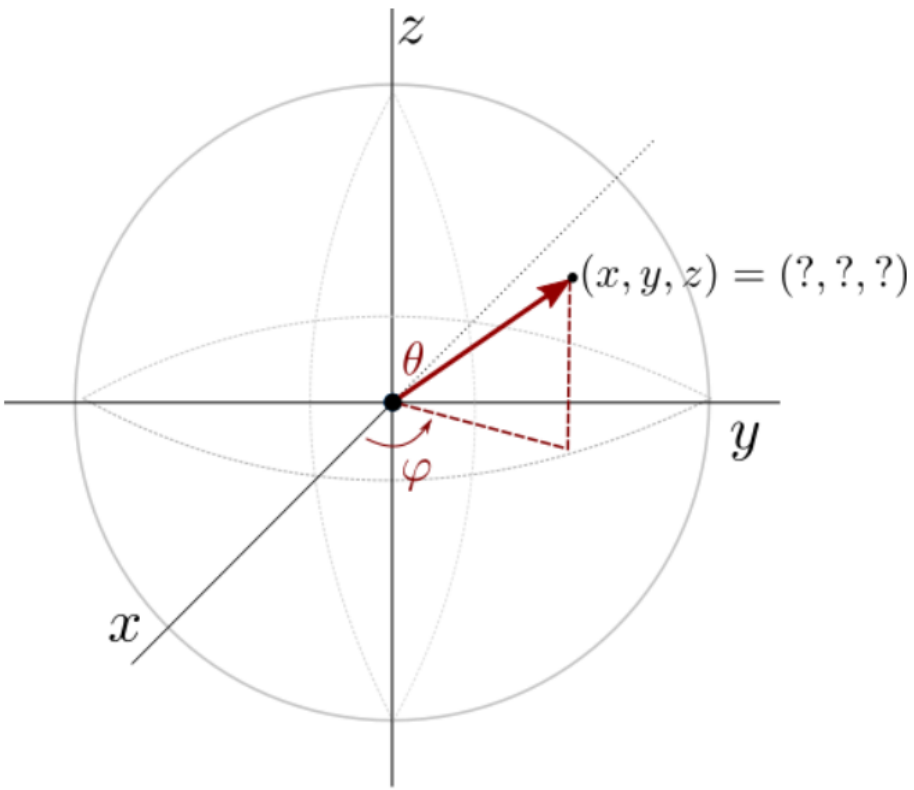
What about expectation values? For pure states $|\psi\rangle$,

For a mixed state,

Mixed states and the Bloch sphere

Practice problem set 4...

Write a function that uses quantum measurements to extract its Bloch vector, i.e., its coordinates in 3-dimensional space, as demonstrated by the following graphic:



Mixed states and the Bloch sphere

Given $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$, can we determine ρ?

Recall ρ is Hermitian; Paulis are a basis.

ρ must have trace 1:

Trace out another Pauli:

Mixed states and the Bloch sphere

More formally, we can write any ρ as

where $a_P = \mathrm{Tr}(P\rho) = \langle P \rangle$.

(Note this also generalizes to multi-qubit case)

Mixed states and the Bloch sphere

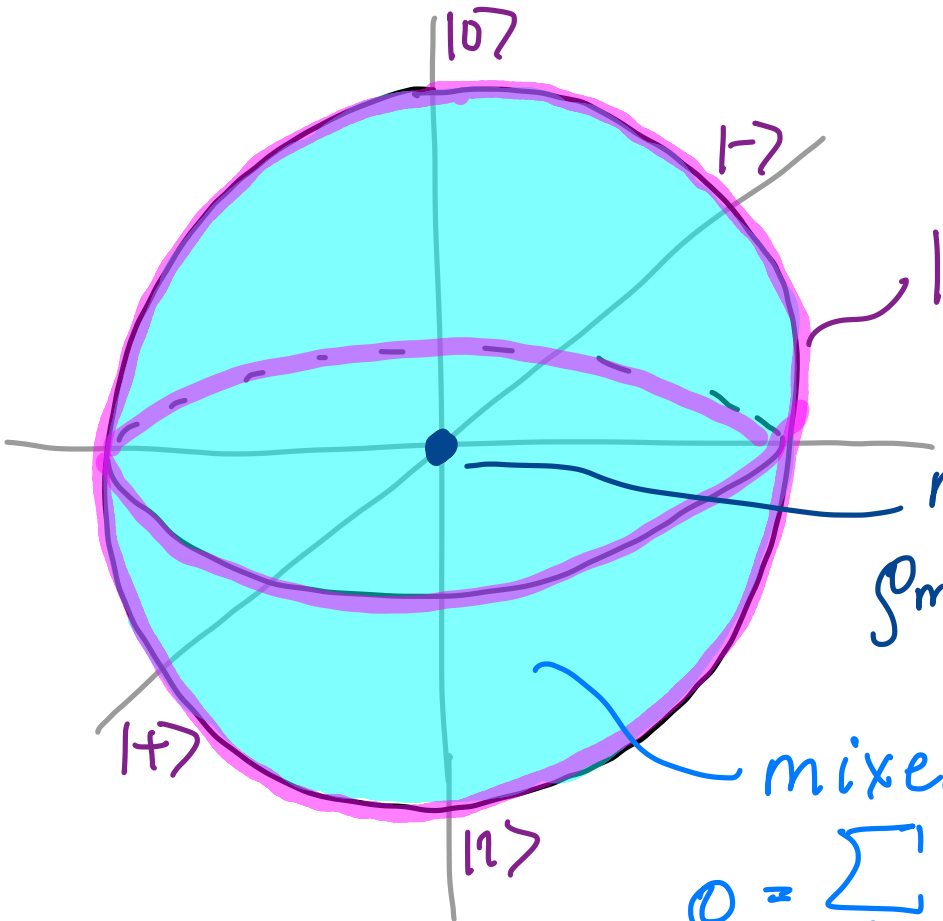
Re-express:

Exercise: As ρ is positive semidefinite, its eigvals are ≥ 0. What constraint does this put on $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$? (Hint: look at $\det(\rho)$)

Mixed states and the Bloch sphere

Exercise: How many eigenvalues does ρ have if it is pure? What constraint does this put on $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$?

Mixed states live *in* the Bloch sphere!



pure states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

maximally mixed state

$\rho_{max} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{1}$

mixed states

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

Mixed states and the Bloch sphere

Exercise: What happens to a mixed state on the Bloch sphere if we apply a unitary U?

Quantum channels

To "get inside" the Bloch sphere, we have to apply a different kind of operation: a quantum channel.

A quantum channel Φ *maps* density matrices to density matrices.

Quantum channels are completely positive, trace-preserving (CPTP) linear maps.

- Trace-preserving:
- Positive:
- Completely positive:

Quantum channels

Channels are characterized by a set of **Kraus operators** $\{K_i\}$,

where

A channel's Kraus operators represent, loosely, a set of things that can happen to a system, including *measurement*, and *errors*.

Example: a channel with one Kraus operator is a unitary channel.

Example: measurement

Recall the general POVM defined earlier: let $\{M_m\}$ be a set of measurement operators and $E_m = M_m^\dagger M_m$,

The $\{M_m\}$ can be viewed as Kraus operators:

Example: measurement

Consider arbitrary mixed state

What is the probability of measuring and obtaining outcome m?

Example: measurement

(Normalized) state after measurement and obtaining outcome m:

Overall state transforms as

Example: measurement

Concrete example: $\{M_m\} = \{\Pi_+, \Pi_-\}$:

Consider mixed state

Intuitively, after measurement expect

Next time

Content:

- Error channels
- Noise in quantum systems

Action items:

1. Work on project (fill out surveys!)
2. Prepare for quiz 8

Recommended reading:

- From this class: Codebook NT; N & C 2.2.6, 2.4
- For next class: Codebook module NT, D; N & C 8.2-8.3,